

9/11 Ch. 1.2

$f \circ g(x) = f(g(x))$: composition of $f(\cdot)$ and $g(\cdot)$

positive measure: 正向角 (逆时针)

negative measure: 负向角 (顺时针)

adjacent (邻边): x -coordinate

$$\sin = \frac{y}{h}$$

$$\csc = \frac{h}{y}$$

opposite (对边): y -coordinate

$$\cos = \frac{x}{h}$$

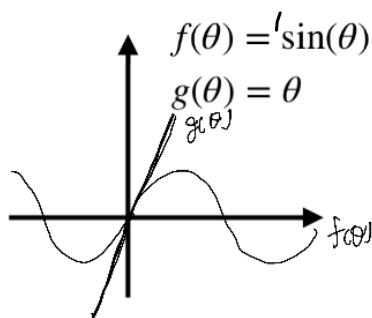
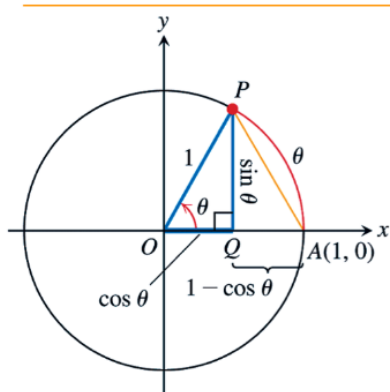
$$\sec = \frac{h}{x}$$

hypotenuse (斜边): distance of a point to origin

$$\tan = \frac{y}{x}$$

$$\cot = \frac{x}{y}$$

Characteristic of $\sin(\theta)$



- Consider a unit circle (with radius 1)
- The arc length of the central angle $0 \leq \theta \leq \pi/2$ is previously obtained as

$$\widehat{PA} = \theta$$

$$\sin^2 \theta + 1 - 2 \sin \theta \cos \theta + \cos^2 \theta$$

$$2 - 2 \sin \theta \cos \theta$$

- In addition,

$$PA \geq \overline{PA} = \sqrt{\sin^2 \theta + (1 - \cos \theta)^2} \geq \sqrt{\sin^2 \theta} = \sin \theta$$

- We thus conclude that

$$\sin(\theta) \leq \theta \text{ for } 0 \leq \theta \leq \pi/2$$

21

Exponential functions: 指数函数: $f(x) = c \cdot a^x$

Inverse functions: A function can be inferred its input by its output.

One-to-One functions: Every output results from uniquely one input, then the answer is positive.
For any two distinct input x_1 and x_2 We have $h(x_1) \neq h(x_2)$

Inverse

Definition. If $h(\cdot)$ is one-to-one, then the inverse of $h(\cdot)$ exists and is defined by $h^{-1}(\cdot)$

Inverse graph

The graphs of $g(x)$ and $g^{-1}(x)$ are reflections of each other about the line $y = x$!

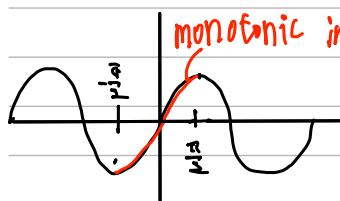
Logarithm function

Domain restriction: One which makes Trigonometric functions one to one

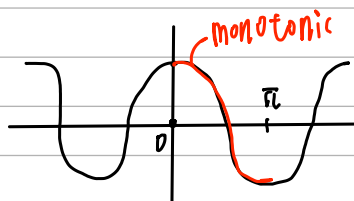
$$\sin \theta : \text{in } -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$$

$$\cos \theta : \text{in } 0 \leq \theta \leq \pi$$

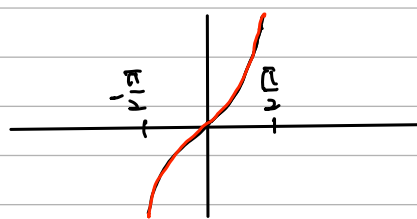
$$\tan \theta : \text{in } -\frac{\pi}{2} < \theta < \frac{\pi}{2}$$



Domain: $[-1, 1]$



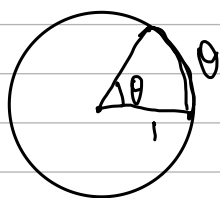
Domain: $[-1, 1]$



Domain: $(-\infty, \infty)$

Inverse of Trigonometric Functions

The inverse of (the properly restricted) $\sin x$ is defined as $\sin^{-1}x = \arcsin x$



$$\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$$

$$\theta = \frac{\pi}{3}$$

$\arcsin$: The arc whose $\sin x$ is $\frac{\sqrt{3}}{2}$

same thing:

The inverse of (the properly restricted) $\cos x$ is defined as $\cos^{-1}x = \arccos x$

Relation between the inverse of trigonometric functions.

$$\cos^{-1}x + \sin^{-1}x = \frac{\pi}{2}$$

$$\cot^{-1}x + \tan^{-1}x = \frac{\pi}{2}$$

$$\csc^{-1}x + \sec^{-1}x = \frac{\pi}{2}$$

* D & R of $\csc^{-1}$ $\sec^{-1}$ $\cot^{-1}$

$$\csc^{-1}: D: (-\infty, -1) \cup (1, \infty) \quad / \quad R: [-\frac{\pi}{2}, 0) \cup (\frac{\pi}{2}, \pi]$$

$$\sec^{-1}: D: (-\infty, -1] \cup [1, \infty) \quad / \quad R: [0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$$

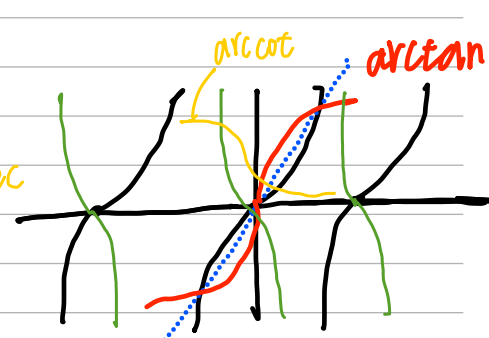
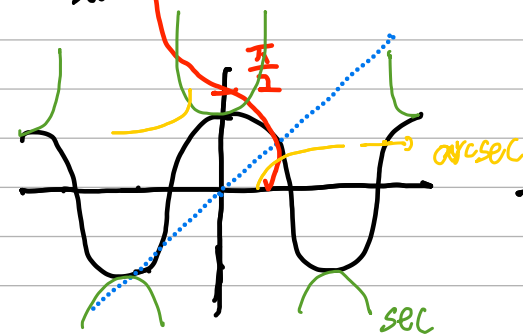
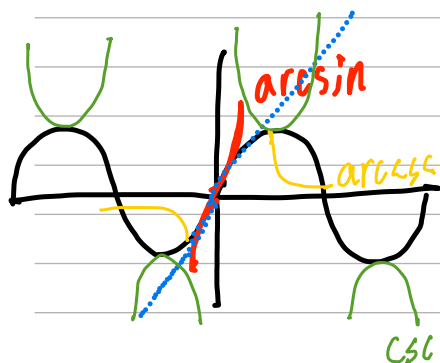
$$\cot^{-1}: D: (-\infty, \infty) \quad / \quad R: (0, \pi)$$

Graph of trigonometric function

$\sin/\sin^{-1}$ $\csc/\csc^{-1}$

$\cos/\cos^{-1}$ $\arccos$
 $\sec/\sec^{-1}$

$\tan/\tan^{-1}$ $\cot/\cot^{-1}$



Ch.2.2

Identity Function: Directly output the result from input

Unit Step Function:
$$\begin{cases} 1 & , x < 0 \\ \frac{1}{2} & , x = 0 \\ 1 & , x > 0 \end{cases}$$